(b) By part (a), the plane in (1) is

(1) $$\xi_1 x_1 + \dots + \xi_n x_n - t = c,$$

So the wave sol-ns

$$u(x_1, x_2, \dots, x_n, t) = F(\xi_1 x_1 + \xi_2 x_2 + \dots + \xi_n x_n - t) \overset{\text{by (1)}}{=} F(c) \Rightarrow$$

(8) $$u = F(c) = \text{const},$$

establishing the desired result. ∎

Prob. 8. (2&T, Ch. 8. 1.6)

Rewrite the spherical wave eq-n (why is this called the spherical wave eq-n?):

(1) $$\frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \cdot \frac{\partial u}{\partial r}\right] - \frac{\partial^2 u}{\partial t^2} = 0, \qquad (n=3)$$

as a one dim-l wave eq-n: $w_{rr} - w_{tt} = 0$ using the change of <u>dependent</u> variables $w = ru$.

Sol-n: [Preamble].

Consider the ↓ change of variables for dep-t var-le:

(2) $$w = r \cdot u \Rightarrow \boxed{u = \frac{w}{r}}, \quad \boxed{w = w(r,t)}.$$ Then

(3) $$\frac{\partial u}{\partial r} = \frac{\partial}{\partial r}\left[\frac{1}{r} \cdot w\right] = \frac{-1}{r^2} w + \frac{1}{r} \cdot w_r \quad \&$$

(4) $$\frac{\partial u}{\partial t} = \frac{\partial}{\partial t}\left[\frac{1}{r} \cdot w\right] = \frac{1}{r} w_t$$

Putting this into the LHS of (1) gives

$$\text{LHS} = \frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \cdot \left(-\frac{1}{r^2} w + \frac{1}{r} \cdot w_r\right)\right] - \frac{\partial}{\partial t}\left[\frac{1}{r} \cdot w_t\right] =$$

$$= \frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[-w + r \cdot w_r\right] - \frac{1}{r} \cdot w_{tt} =$$

$$= \frac{1}{r^2}\left[-w_r + \frac{\partial}{\partial r}\left[r \cdot w_r\right]\right] - \frac{1}{r} w_{tt} = \frac{1}{r^2} \cdot \left[-\cancel{w_r} + \cancel{w_r} + r \cdot w_{rr}\right] - \frac{1}{r} \cdot w_{tt}$$

$$= \frac{r \cdot w_{rr}}{r^2} - \frac{1}{r} \cdot w_{tt} = \frac{1}{r}\left[w_{rr} - w_{tt}\right].$$

So, (1) becomes.

$$\frac{1}{r}\left[w_{rr} - w_{tt}\right] = 0 \quad \Rightarrow$$

(5) $$\boxed{w_{rr} - w_{tt} = 0}$$

as desired.

- why (1) is called spherical <u>wave eq-n</u>?

By the def-n, a spherical wave is a sol-n of the wave in the x-space centered at the origin. Laplacian operator in spherical coor-s:

(6) $$\frac{1}{r^{n-1}} \frac{\partial}{\partial r}\left(r^{n-1} \cdot \frac{\partial u}{\partial r}\right) + \frac{1}{r^2} \Lambda_n u - \frac{\partial^2 u}{\partial t} = 0.$$

Yet, by def-n, a sprerical wave is a sol-n of (6) which only depends on r & t, does not depend on angular var-s, hence

$$\Lambda_n u = 0$$

having the form in (1) for $n=3$ case, in turn.

$\blacksquare$